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Brain tumor classification for MR images using transfer learning and fine-tuning

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ABSTRACT

Accurate and precise brain tumor MR images classification plays important role in clinical diagnosis and decision making for patient treatment. The key challenge in MR images classification is the semantic gap between the low-level visual information captured by the MRI machine and the high-level information perceived by the human evaluator. The traditional machine learning techniques for classification focus only on low-level or high-level features, use some handcrafted features to reduce this gap and require good feature extraction and classification methods. Recent development on deep learning has shown great progress and deep convolution neural networks (CNNs) have succeeded in the images classification task. Deep learning is very powerful for feature representation that can depict low-level and high-level information completely and embed the phase of feature extraction and classification into self-learning but require large training dataset in general. For most of the medical imaging scenario, the training datasets are small, therefore, it is a challenging task to apply the deep learning and train CNN from scratch on the small dataset. Aiming this problem, we use pre-trained deep CNN model and propose a block-wise fine-tuning strategy based on transfer learning. The proposed method is evaluated on T1-weighted contrast-enhanced magnetic resonance images (CE-MRI) benchmark dataset. Our method is more generic as it does not use any handcrafted features, requires minimal preprocessing and can achieve average accuracy of 94.82% under five-fold cross-validation. We compare our results not only with the traditional machine learning but also with deep learning methods using CNNs. Experimental results show that our proposed method outperforms state-of-the-art classification on the CE-MRI dataset.

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1. Introduction and related works

Timely diagnosis of brain tumor plays important role in effective treatment planning and patient care. Manual classification of brain tumor MR images with similar structures or appearances is a difficult and challenging task, which depends on the availability and expertise of the radiologist to identify the brain tumor and classify. The radiologist performs two types of brain tumor classification: (i) identify whether the brain MR images are normal or abnormal. (ii) Classify the abnormal brain MR images into different tumor types. Such manual brain tumor classification is impractical, nonreproducible and time intensive for large amount of MRI data. To address this problem, automatic classification is a possible solution to classify brain tumor MR images with minimum intervention by radiologists. In this research, we focus on classifying abnormal brain (tumor) images. The CE-MRI dataset (Cheng, 2017) utilized in this study consists of three types of brain tumors with the highest percentage among brain tumors. In clinical practice, the incident rates of glioma, meningioma, and pituitary tumor are approximately 45%, 15%, and 15%, respectively, among all brain tumors. Compared with the simple binary classification (normal and abnormal), our task is more complex and challenging.

Traditional machine learning methods for classification usually consist of several steps, including preprocessing, feature extraction, feature selection, dimension reduction, and classification. The feature extraction usually depends on the expert's knowledge in the specific domain. It is a challenging task for a non-expert to use traditional machine learning methods in their research. Feature extraction is a core step in traditional machine learning and classification accuracy depends on the extracted features. Feature extraction can be categorized into two types. The first type is low-level (global) feature, such as intensity and texture features,

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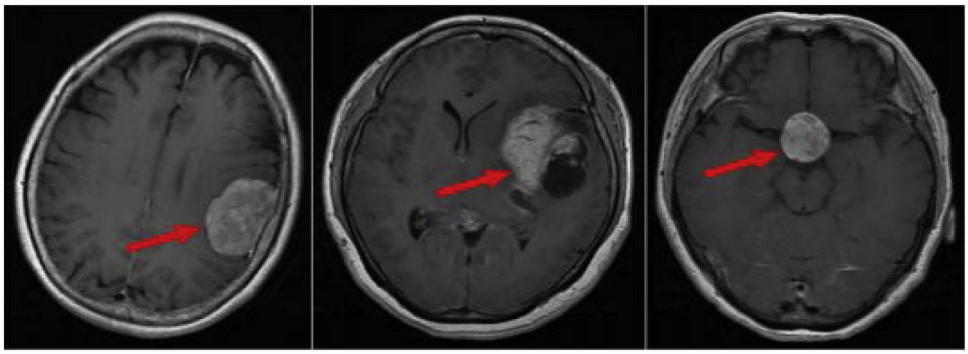


Fig. 1. (a) and (b) are meningiomas with different appearance, and (c) is a pituitary tumor with an appearance similar to that of (a).

first-order statistics (e.g., mean, standard deviation and skewness) and second-order statistics derived from gray level co-occurrence matrix (GLMC), shape, wavelet transform, and Gabor feature. Jiang et al. (2013) segmented brain tumor based on voxel classification by AdaBoost classifier using graph cut and Gabor features. Javed et al. (2013) proposed multiclass classification method using texture features, fuzzy weighting, and support vector machine (SVM). Selvaraj et al. (2007) used first-order and second-order statistics and built a binary classifier using least square SVM for the classification of normal and abnormal brain MR images. John (2012) proposed discrete wavelet transformation and GLCM based method for tumor identification and classification. The low-level features can efficiently describe the image, but these features are low-level, and their representation power is limited because different types of brain tumors have a similar appearance and the same type varies in appearances such as boundary, texture, size, and shape, as shown in Fig. 1.

The second type is high-level (local) feature, such as bag-of-words (BoW), fisher vector (FV), and scale-invariant feature transformation (SIFT). Many methods used BoW for medical image classification and retrieval. Such as X-ray images classification and retrieval on organ and pathology levels (Avni et al., 2011), classification of breast tissue density in mammograms (Bosch et al., 2006), content-based retrieval of brain tumor (Yang et al., 2012a) and liver lesions (Yang et al., 2012b). Cheng et al. (2016) proposed brain tumor retrieval using FV. Lazebnik et al. (2006) used spatial pyramid matching, divided the image into rectangular grids, computed BoW descriptor for each grid, and finally weighted and combined the results. The spatial features calculated from a symmetrical rectangular region is not suitable for the brain tumor partition due to large geometric transformation (such as translation and rotation) and variation in the appearance of the tumor (e.g., intensity, size, and shape). Cheng et al. (2015) proposed specialized brain tumor classification for CE-MRI dataset using augmented tumor region of interest (ROI), image dilation and ring-form partition. They computed Intensity histogram, GLMC, and BoW features and achieved the accuracy of 91.28%. The statistical feature extracted from BoW, FV, and SIFT are high-level features and formulated on local scale that certainly ignores spatial information.

There are two problems in the feature extraction phase of traditional machine learning. First, it focuses only on either low-level or high-level features. Second, it depends on handcrafted features. For example, the CE-MRI dataset, the content of a specific category is distributed with intrinsic irregularity. There is a strong correlation between the layout of the tumor, edema, and surrounding normal tissues. Meningioma and pituitary tumor are similar in shape as shown in Fig. 2(a), 2(b). Meningioma is generally adjacent to the skull, gray matter, and cerebrospinal fluid. A pituitary tumor is near the sphenoidal sinus, internal carotid arteries, and optic chiasma. In appearance, glioma is dissimilar in shape and generally surrounded by edema as shown in Fig. 2(c) and 2(d). Therefore, the most important information and the discriminative features of brain tumors are related to the location/position of the tumor region in the MR image together with its boundary, texture, size, and shape. Cheng et al. (2015) uses handcrafted features (i.e., ring-form partition of tumor region) which require strong prior information(i.e., the position or location of the tumor in an image); thus, it is not a simple task, prone to human errors and might cause inter- and intra-operator deviations (Jiang et al., 2013). Accordingly, it is necessary to design such a method to encode/combine both low-level and high-level features without using handcrafted features.

Recent studies (Greenspan et al., 2016; Litjens et al., 2017; Ren et al., 2017) have proved that deep learning approaches do not need manually extracted (handcrafted) features. Deep learning embeds the phase of feature extraction and classification in self-learning. It needs a dataset with only minimal preprocessing if required and then determines significant features in a self-learning manner (Bengio, 2009; LeCun et al., 2015). A key challenge in MR image classification is to reduce the semantic gap between the low-level visual information captured by the MRI machine and the high-level semantic information perceived by the human evaluator. For classification, the effectiveness of such a method is more important in terms of feature representations that can depict low-level and high-level information completely without handcrafted features. To justify the feasibility of the proposed method, CNNs automatically generate powerful discriminative features using a hierarchical learning approach. Feature maps in the earlier layers extract low-level features, and feature maps in higher layers extract high-level domain (content) specific features. Earlier layers feature maps encode simple structural information such as edges, shape, and textures, and higher layers combine these low-level feature maps to encode/construct abstract representation, which integrates local and global information. Thus, deep learning can covert the manual feature extraction into self-learning and facilitate non-expert of machine learning to use it efficiently.

Empirically, the deep neural network learns the high-level abstraction from the low-level abstraction in a hierarchical fashion. Deep learning outperformed state-of-the-art methods in various field of machine learning such as natural language processing (Collobert and Weston, 2008), generating text (Sutskever et al., 2011), speech recognition (Hinton et al., 2012), object detection (Szegedy et al., 2013), face verification (Taigman et al., 2014), image description (Karpathy and Fei-Fei, 2015), the game of Go (Silver et al., 2016), machine translation (Zhang and Zong, 2015), and grand challenges (Everingham et al., 2011; Roux, 2014; Russakovsky et al., 2015). In particular, enhanced performance in computer vision

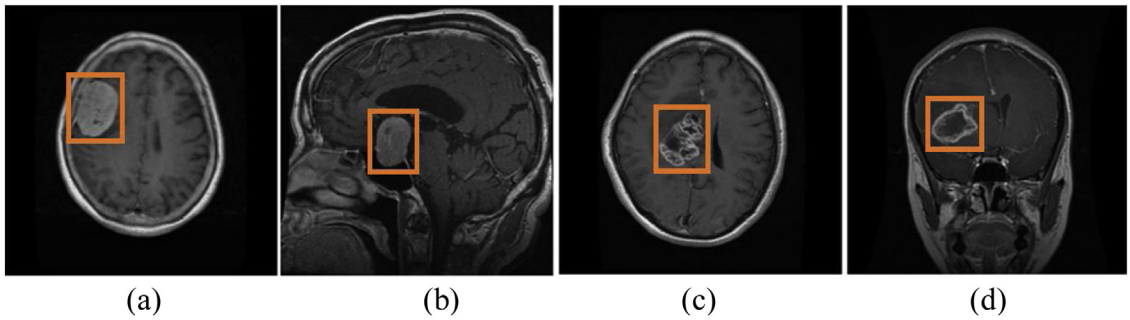


Fig. 2. Four images of brain tumors in T1-weighted CE-MRI. The region inside the red rectangle contains a tumor. (a) Meningioma located near the skull, (b) pituitary tumor located near the sphenoidal sinus, (c) glioma containing edema and necrosis, and (c) glioma surrounded by edema.

encouraged the use of deep learning in medical image analysis, such as classification, detection and segmentation (Kleesiek et al., 2016; Tajbakhsh et al., 2016; Zhang et al., 2015), fusion (Suk et al., 2014), computer-based diagnosis and prediction (Suk et al., 2015, 2016), lesion/landmark detection (Dou et al., 2016; Pereira et al., 2016; van Tulder and de Bruijne, 2016), and microscopic image examination (Chen et al., 2016; Cireşan et al., 2013).

CNNs have been used for decades but were not popular until Krizhevsky et al. (2012) employed the deep learning approach (AlexNet) and won the ImageNet Large-Scale Visual Recognition Challenge (ILSVRC) in 2012. Simonyan and Zisserman (2014) introduced a similar but deeper CNN (VGG Net) secured first place in the localization task and second place in the classification task in the ILSVRC in 2014. Deep learning, and especially CNNs, got success due to the advancement of computational technologies such as powerful GPUs, the development of learning algorithms (Hinton et al., 2006; Hinton and Salakhutdinov, 2006; Ioffe and Szegedy, 2015; Nair and Hinton, 2010; Srivastava et al., 2014), and the availability of big data (Everingham et al., 2011; Roux, 2014; Russakovsky et al., 2015).

Recent years, some researchers used CNNs for brain tumor MR images classification on CE-MRI dataset (Abiwinanda et al., 2019; Paul et al., 2017). Abiwinanda et al. (2019) proposed a simple CNN architecture that consists of one convolution, max-pooling, and flattening layers, followed by a full connection layer, trained from scratch and achieved the classification accuracy of 84.19%. Paul et al. (2017) trained four layers CNNs (consisting of two convolutional and two fully connected layers) from scratch and achieved the classification accuracy to 91.43%, however, the accuracy of image-based is 90.26%, which is lower than Cheng's-90.28%. Abiwinanda et al. (2019) and Paul et al. (2017) proposed CNNs training-validation loss and accuracy have a large gap. The training-validation progress and results show that their CNNs suffered from overfitting because of the small sample size of the CE-MRI dataset.

CNNs have shown good performance in computer vision on large labeled datasets, such as ImageNet (Russakovsky et al., 2015), which contains more than one million labeled images. However, it is difficult to apply such deep CNNs in the medical imaging field. First, the sample size of the medical image datasets is usually small because such datasets require the availability of expert radiologists to manually examine and label the images, which is time-consuming, laborious and costly. Second, training deep CNN is a complicated task for a small dataset because of overfitting and convergence problem. Third, domain expertise is required to repeatedly revise the model and adjust the learning parameters and hyperparameters of the model to achieve better performance. Therefore, training deep CNN from scratch is a challenging task that is tedious and demands much diligence and patience. For the small dataset scenario, a favorable substitute for training CNN from scratch is to use a pre-trained CNN based on transfer learning and fine-tuning strategy (Shin et al., 2016, 2017; Tajbakhsh et al., 2016).

Pre-trained CNN models have been effectively used to several computer vision (Azizpour et al., 2015; Penatti et al., 2015; Sharif Razavian et al., 2014) and medical imaging applications (Shin et al., 2016, 2017; Tajbakhsh et al., 2016) as feature extractors or as a baseline for transfer learning. The main advantage of CNNs is the "transferability" of the knowledge learned in pre-trained models. In the literature, two types of transfer learning approaches have been proposed for medical imaging, such as (i) Use off-the-shelf trained CNNs model over a large dataset of natural images, extract features from a specified layer of a pre-trained model for the medical imaging dataset and train a separate learning method for classification (Ciompi et al., 2015; Shin et al., 2016). Bar et al. (2015) used pre-trained CNN as a feature extractor for chest pathology identification. van Ginneken et al. (2015) extracted features from pre-trained CNN, integrated handcrafted features with CNN-based features for the detection of a pulmonary nodule in CT scan. (ii) Use pre-trained CNNs and apply fine-tuning to the application of medical images database. For the instance, Carneiro et al. (2015) produced promising results by replacing the fully connected layer of pre-trained CNN with a classifier layer, kept the rest of the CNN same and trained only the classifier layer for classification of unregistered multi-view mammogram. Chen et al. (2015) proposed fine-tuning of pre-trained CNN for localizing standard planes in ultrasound images. Gao et al. (2016) fine-tuned all layers of pre-trained CNN for automatic classification of interstitial lung diseases. Shin et al. (2015) used pre-trained CNN and fine-tuned for automatic mapping of medical images to different document-level. Margeta et al. (2017) fine-tuned pre-trained CNN for automatic retrieval of missing or noisy cardiac acquisition plane information from MRI and predicted five most common cardiac views.

In this research, we proposed an automatic classification system for multiclass brain tumor MR images, which is more complex and challenging task than simple binary classification. The CE-MRI dataset (Cheng, 2017) is very small as compared with natural images datasets (Everingham et al., 2011; Russakovsky et al., 2015). It is a challenging task for the small dataset to train deep CNN from scratch with proper convergence and without suffering from overfitting (Shin et al., 2016, 2017; Tajbakhsh et al., 2016). Inspired from other researcher's work (Azizpour et al., 2015; Penatti et al., 2015; Sharif Razavian et al., 2014; Shin et al., 2016, 2017; Tajbakhsh et al., 2016), we adopt the idea of transfer learning from natural images. Our method is different from (i) and (ii) discussed above. We used pre-trained CNN VGG19, which is trained on a large dataset of ImageNet (Simonyan and Zisserman, 2014) and adopt the following strategy.

1 The weights of CNN are initialized with pre-trained VGG19
2 To achieve enhance classification results, we propose block-wise fine-tuning, gradually goes deep down into earlier blocks of the CNN, and monitor the performance improvement on CE-MRI dataset. Our proposed block-wise fine-tuning strategy is not sim-

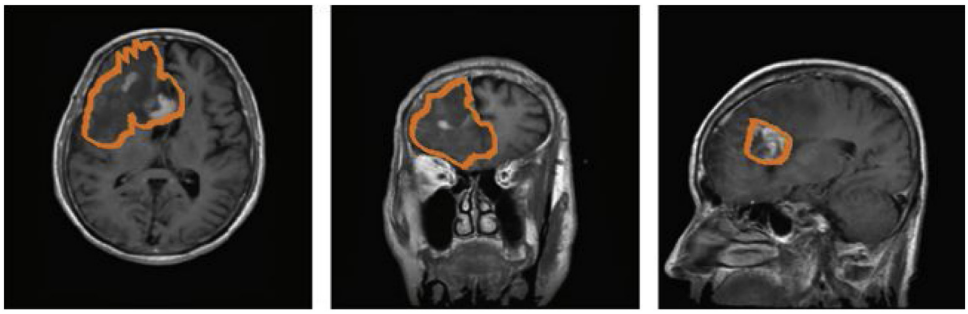


Fig. 3. Three views of glioma in a patient. The area inside the yellow outline indicates a tumor.

ilar to those work (Carneiro et al., 2015; Chen et al., 2015; Gao et al., 2016; Margeta et al., 2017; Shin et al., 2015), in which the complete pre-trained CNN is fine-tuned at once.

3 The proposed strategy is different from (Bar et al., 2015; Ciompi et al., 2015; van Ginneken et al., 2015), in which pre-trained CNNs is used as off-the-shelf feature extractor and trained a separate classifier for classification.

The layout of this paper is organized as follows: The proposed research methodology is given in section 2. The experimental settings, parameter optimization, performance metrics, results analysis and comparison are shown in Section 3. A brief discussion is provided in Section 4, and the conclusion and future work is presented in Section 5.

2. Proposed method

2.1. Dataset

In this research, we used a publicly available CE-MRI dataset (Cheng, 2017) available at (https://figshare.com/articles/brain_tumor_dataset/1512427). The proposed brain tumor classification is based on two-dimensional images (2D slices), not 3-D volume, because in most clinical practice, the acquired and available CE-MRI images are 2-D slices with a large slice gape. Therefore, our classification system based on 2-D MR images for clinical application is practical. The dataset was collected during 2005–2010 from Nanfang Hospital, Guangzhou, China, and General Hospital, Tianjin Medical University, China. The dataset contains three types of tumors (i.e., glioma, meningioma, and pituitary tumor as shown in Fig. 1 and 2) from 233 patients with total 3064 images across the axial, coronal, and sagittal views, as shown in Fig. 3. Table 1 shows the details of the CE-MRI dataset. The images in the dataset are provided in matrix form. The size of each image is 512 × 512 pixels, and the pixel size is 49 mm × 49 mm.

2.2. Preprocessing

The T1-weighted CE-MRI data are 2-D images of size 512 × 512, as shown in Fig. 4(a). In this work, we provided MR images directly to the CNN and the convolutional kernel is applied to the pixel intensity in the image. The result of a convolutional kernel relies heavily on these intensity values. However, intensity values in MR images do not have a fixed meaning, and it has been observed that intensity values across MR images vary greatly within or between subjects. These intensity values of MR images are also sensitive to the acquisition condition. Data mining and especially CNN approaches need to normalize the inputs; otherwise, the network will be ill-conditioned. In principle, normalization is performed to obtain the same range of values for each input into the CNN model, which can guarantee a stable convergence. In this scenario, intensity normalization is necessary to preprocess MR images. Therefore, we normalize the CE-MR images by using min-max normalization to scale the intensity value to [0, 1], which is computed as follows:

$$y_i = (x_i - min(x))/(max(x) - min(x)) \quad (1)$$

where y_i is the normalized intensity value against position x_i (where $i = 1. . ..n$) and *min(x)* and *max(x)* is the minimum and maximum intensity values, respectively, across the entire image. The normalized CE-MR image is shown in Fig. 4(b). After normalization, we resize the normalized image to 224 × 224 and duplicate it three times to create three channels according to the input size of the pre-trained VGG19.

The intensity normalization brings the intensity values within a coherent range across all the MR images and facilitates learning in the training process. Resizing the images speeds up the training process and reduce the memory requirement.

2.3. Training convolutional neural networks (CNNs)

The training of CNNs starts from the first input layer to the final classification layer in a feed-forward fashion; and then, error back-propagation starts from the classification layer towards the first convolutional layer. Neuron i in layer l receives input from neuron j of layer l-1 in a forward pass computed as follows:

$$In_i^l = \sum_{j=1}^{n} W_{ij}^l x_j + b_i \quad (2)$$

The output is computed by a nonlinearity ReLu function:

$$out_i^l = max(0, In_i^l) \quad (3)$$

All neurons in the convolutional and fully connected layers use Eqs. (2) and (3) to calculate the input and produce output in the form of nonlinear activation. The pooling layer uses a $K \times K$ square window sliding on the $N \times N$ feature map and takes the maximum or average value of the features inside the window. Thus, it decreases the spatial size of the feature map from $N \times N$ to $\frac{N}{K} \times \frac{N}{K}$.

The final layer computes the classification probability of each tumor type using the Softmax function:

$$out_i^l = \frac{e^{In_i^l}}{\sum_i e^{out_k^l}} \quad (4)$$

CNNs are trained with the back-propagation algorithm by minimizing the following cost function with respect to the unknown weights W:

$$C = -\frac{1}{m}\sum_{i}^{m} \ln\ (p\ (y^{\ i}|X^{\ i})) \quad (5)$$

Table 1
Details of the CE-MRI dataset.

Tumor type	Number of patients	Number of MR images	MRI View	Number of MR images
Meningioma	82	708	axial	209
			coronal	268
			sagittal	231
Glioma	89	1426	axial	494
			coronal	437
			sagittal	495
Pituitary	62	930	axial	291
			coronal	319
			sagittal	320
Total	233	3064		3064

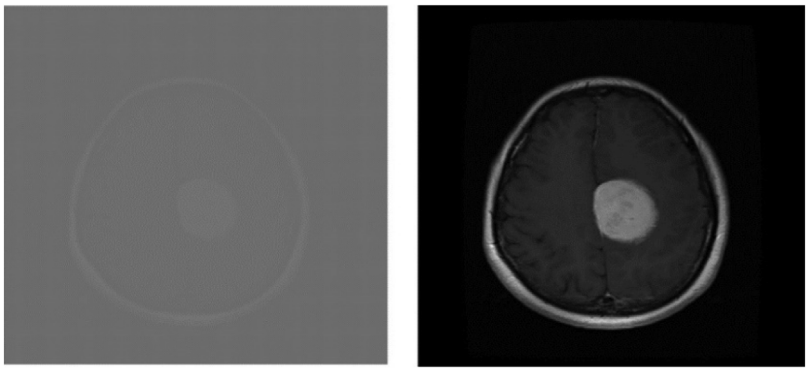


Fig. 4. (a) CE-MRI and (b) normalized CE-MRI.

where m is the total number of training samples in the training set, X^i is the ith sample in the training set with the label y^i and $p\,(y^i|X^i)$ is the true classification probability. The cost function C is minimized using stochastic gradient descent over the mini-batches of size N and the training cost is approximated by the mini-batch cost. Considering that W_l^t represents the weights at iteration t for convolutional layer l and $\hat{C}$ represents the mini-batch cost, we then update the weights in the next iteration as follows:

$$\gamma^t = \gamma^{\lfloor tN/m \rfloor}$$

$$V_l^{t+1} = \mu V_l^t - \gamma^t \alpha_l \frac{\partial \hat{C}}{\partial W_l}$$

$$W_l^{t+1} = W_l^t + V_l^{t+1} \tag{6}$$

where a_l is the learning rate of layer l, γ is the scheduling rate that decreases the initial learning rate α at the end of a specified number of epochs, and μ is the momentum that describes the influence of previously updated weights in the current iteration.

2.4. Transfer learning and fine-tuning of CNN (VGG19)

During the training process, the weights of the CNN layers are updated after every iteration by Eq. (6). There are 19 layers and 144 million trainable parameters (weights) in the VGG19 architecture. For the training and optimization of such deep network, the large dataset is necessary. However, for small dataset, it is very difficult to determine the appropriate local minima for the cost function in Eq. (5), and the network will suffer from overfitting. Therefore, we initialized weights from the pre-trained VGG19 model.

After the weights transfer, we fine-tuned VGG19 on CE-MRI dataset. The VGG19 consists of sixteen convolutional layers and three fully connected layers, as shown in Table 2. If we apply the strategy of layer-wise fine-tuning by adding one layer each time, then it will need to fine-tune nineteen layers. Because five-fold cross-validation is under consideration, it will need to fine-tune ninety-five VGG19 architectures. Suppose that the time of training of each architecture is approximately thirty minutes, then it will take more than one week to complete the fine-tuning of the VGG19 in a layer-wise manner. Similarly, determining the optimal parameters for the layer-wise fine-tuning will be more time-intensive. Small improvement in the results was observed when adopting a layer-wise fine-tuning approach. Therefore, the VGG19 architecture is divided into six blocks based on pooling layers, as shown in Table 2. The block-wise architecture of the VGG19 is shown in Fig. 5. The final fully connected layer of VGG19 is composed of 1000 neurons corresponds to classes in the ImageNet dataset, so, the final fully connected layer here is changed to three neurons according to classes in the CE-MRI dataset.

The deep CNN is trained in a block-wise approach by starting fine-tuning from the final block and keeping all other blocks (layers) fixed by freezing their learning. Suppose B is the total number of blocks, α_b is the learning rate of block b, and we want to fine-tune only the final block (B) and then set $\alpha_b = 0$ for all blocks except block B. If fine-tuning the last two blocks, then set $\alpha_b = 0$ for b $\neq$ B, B-1. Similarly, set the learning parameters of all blocks for fine-tuning as shown in Table 3.

Earlier layers in the pre-trained CNNs contain the generic feature, and later layers contain the domain (content) specific features of the natural images. The learning of the earlier layers can be frozen because of the low-level features in these layers. To learn the domain-specific features of MRI brain tumors, we start the training by block-wise fine-tuning and initiate from the top block.

Table 2
Pre-trained VGG19 (CNN) Architecture and Parameters.

Block	Layer	Type	Input	Kernel	Stride	Pad	Output
	input	Image Input	$3 \times 224 \times 224$	N/A	N/A	N/A	$3 \times 224 \times 224$
B_1	conv1_1	Convolution	$3 \times 224 \times 224$	3×3	1	1	$64 \times 224 \times 224$
	conv1_2	Convolution	$64 \times 224 \times 224$	3×3	1	1	$64 \times 224 \times 224$
	pool1	Max Pooling	$64 \times 224 \times 224$	2×2	2	0	$64 \times 112 \times 112$
B_2	conv2_1	Convolution	$64 \times 112 \times 112$	3×3	1	1	$128 \times 112 \times 112$
	conv2_2	Convolution	$128 \times 112 \times 112$	3×3	1	1	$128 \times 112 \times 112$
	pool2	Max Pooling	$128 \times 112 \times 112$	2×2	2	0	$128 \times 56 \times 56$
B_3	conv3_1	Convolution	$128 \times 56 \times 56$	3×3	1	1	$256 \times 56 \times 56$
	conv3_2	Convolution	$256 \times 56 \times 56$	3×3	1	1	$256 \times 56 \times 56$
	conv3_3	Convolution	$256 \times 56 \times 56$	3×3	1	1	$256 \times 56 \times 56$
	conv3_4	Convolution	$256 \times 56 \times 56$	3×3	1	1	$256 \times 56 \times 56$
	pool3	Max Pooling	$256 \times 56 \times 56$	2×2	2	0	$512 \times 28 \times 28$
B_4	conv4_1	Convolution	$512 \times 28 \times 28$	3×3	1	1	$512 \times 28 \times 28$
	conv4_2	Convolution	$512 \times 28 \times 28$	3×3	1	1	$512 \times 28 \times 28$
	conv4_3	Convolution	$512 \times 28 \times 28$	3×3	1	1	$512 \times 28 \times 28$
	conv4_4	Convolution	$512 \times 28 \times 28$	3×3	1	1	$512 \times 28 \times 28$
	pool4	Max Pooling	$512 \times 28 \times 28$	2×2	2	0	$512 \times 14 \times 14$
B_5	conv5_1	Convolution	$512 \times 14 \times 14$	3×3	1	1	$512 \times 14 \times 14$
	conv5_2	Convolution	$512 \times 14 \times 14$	3×3	1	1	$512 \times 14 \times 14$
	conv5_3	Convolution	$512 \times 14 \times 14$	3×3	1	1	$512 \times 14 \times 14$
	conv5_4	Convolution	$512 \times 14 \times 14$	3×3	1	1	$512 \times 14 \times 14$
	pool5	Max Pooling	$512 \times 14 \times 14$	2×2	2	0	$512 \times 7 \times 7$
B_6	FC6	Fully Connected	$512 \times 7 \times 7$	7×7	1	0	4096×1
	FC7	Fully Connected	4096×1	1×1	1	0	4096×1
	FC8	Fully Connected	4096×1	1×1	1	0	1000×1

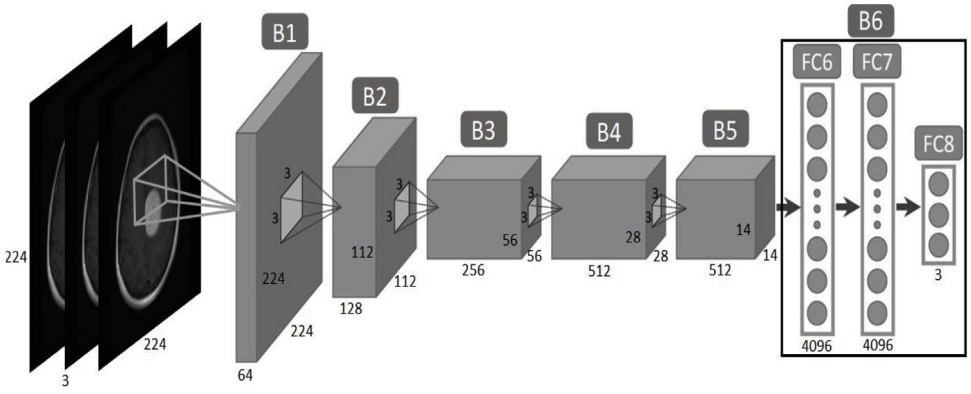


Fig. 5. The proposed Block-Wise VGG19 architecture. B represents the group of layers placed in blocks (shown in Table 2). Final layer replaced with three classes according to classes in the CE-MRI dataset.

Table 3
Fine tuning parameters of the VGG19. FT represents fine-tuning, *b* represents the block, α_b is the learning rate of specified block *b*, and μ, γ, and α are training options.

Fine-tuning Blocks	Training options			Learning rate of each Block (α_b)					
	μ	α	γ	α_{B1}	α_{B2}	α_{B3}	α_{B4}	α_{B5}	α_{B6}
FT:B_1-B_6	0.9	0.01	0.9	0.1	0.1	0.1	0.1	0.1	0.1
FT: B_2-B_6	0.9	0.01	0.9	0	0.1	0.1	0.1	0.1	0.1
FT: B_3-B_6	0.9	0.01	0.9	0	0	0.1	0.1	0.1	0.1
FT: B_4-B_6	0.9	0.01	0.9	0	0	0	0.1	0.1	0.1
FT: B_5-B_6	0.9	0.01	0.9	0	0	0	0	0.1	0.1
FT: B_6	0.9	0.01	0.9	0	0	0	0	0	0.1

3. Experiments and results

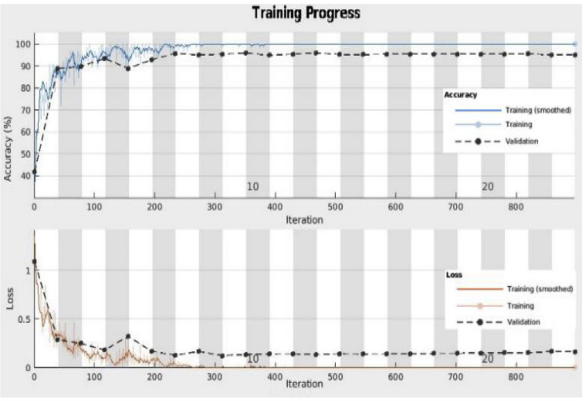
To test the performance of the proposed approach, we adopted the same experimental setup as in (Abiwinanda et al., 2019; Cheng et al., 2015; Paul et al., 2017), and randomly divided the CE-MRI data of 233 patients into five subsets of approximately equal size. We ensured the no overlap and equal ratios of the different types of tumors in the five subsets. We ensured that images from the same patient do not appear simultaneously in the training and testing set. We used five-fold cross-validation to evaluate the performance. In five-fold cross-validation, one subset is used as the test dataset, and the remaining four subsets are sequentially used as the training dataset. The final result is the average classification accuracy of the five-fold test dataset. The proposed research was implemented with MATLAB R2017b and GPU NIVIDA TITAN X (Pascal) with 12 GB onboard memory.

3.1. Training and parameters optimization

The training and block-wise fine-tuning of each CNN take approximately 20–30 minutes, concretely, it depends on the choice of the training and fine-tuning parameters, proper convergence, training-validation accuracy, and error. To find the optimum convergence of each CNN, we properly monitor the improvement of



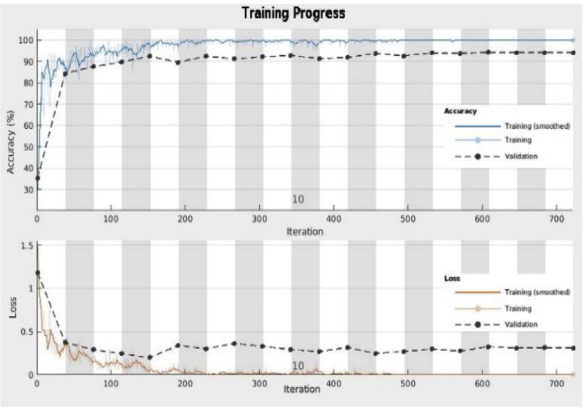
(a). Cross-Validation set1



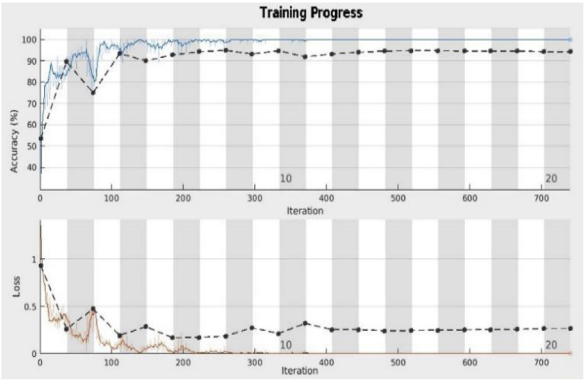
(c). Cross-Validation set3



(b). Cross-Validation set2



(d). Cross-Validation set4



(e). Cross-Validation set5

Fig. 6. Training progress: the Accuracy and the loss history of best performing model FT: B_1-B_6 on fivefold cross-validation. Fig. (a) - (e) represent each set validation and training history.

training-validation accuracy and error. Training will stop automatically if there is no improvement with respect to validation accuracy and error.

Table 3 presents the optimum value of each parameter used in the experiments for fine-tuning. We used trial-and-error-based approach to determine these values and performed experiments with different values of these parameters. During the training process, we found that the proper convergence depends on the initial learning rate α, the learning rate α_b of each layer and the scheduling rate γ. The optimum value for $\alpha = 0.01$ and $\alpha_b = 0.10$ ensured proper convergence. If we set α and α_b very large, then the CNN fails to converge properly and results in low performance on the testing and validation data. If we set α and α_b very small, then the convergence process slows down. The value of γ is related to convergence speed. If the convergence is very slow, then γ should be large enough to keep the learning rate high. If the convergence is very fast, then γ should be small enough to decrease the learning rate and prevent the network from overfitting. Our initial choices of the learning rates α and α_b start the convergence relatively fast. We control α by γ to decrease the learning rate after every five epochs to prevent the network from overfitting. During our experimental investigation, the suitable value of $\gamma = 0.90$. Nesterov's momentum μ describes the influence of previously updated weights in the current iteration, and its most commonly used values are [0.5, 0.9, 0.95 and 0.99]. The reasonable value during our experimental analysis for μ is 0.9.

We set the base-learning rate of each layer twice as the α_b, the mini-batch size for training at 64 (the maximum mini-batch size supported by our GPU for the VGG19) and the maximum epochs at 50 for fine-tuning throughout the experiments. However, most of our fine-tuned CNNs converge between 25 and 35 epochs. The validation test prevents the network from overfitting and helps to monitor proper convergence. We validate the training process after every epoch and stop the training process automatically if there is no improvement on the validation test over 15 epochs. Fig. 6 shows the accuracy and loss of training-validation process for best-

Table 4
Defining the terms TP, FP, FN, TN.

Predicted label	Actual Label	Definition
Positive	Positive	Truly Positive (TP)
Positive	Negative	False Positive (FP)
Negative	Positive	False Negative (FN)
Negative	Negative	True Negative (TN)

performing methods FT: B_1-B_6. We observed from the accuracy and loss graphs that our best performing model FT: B_1-B_6 converge between 20 to 25 epochs.

3.2. Performance metrics

Following the same performance metrics in related references (Abiwinanda et al., 2019; Cheng et al., 2015; Paul et al., 2017), we have evaluated the image-based classification performance based on sensitivity/recall, specificity, precision, F1-score, and accuracy. To evaluate these metrics, we provide some terms first as shown in Table 4, and then define the following mathematical expressions.

$$Sensitivity \quad or \quad Recall = \frac{TP}{TP + F\,N} * 100 \tag{7}$$

$$S\ peci f\ \ icity = \frac{TN}{TN + FP} \ * \ 100 \tag{8}$$

$$Precision = \frac{TP}{TP + FP} \ * \ 100 \tag{9}$$

$$F1 \ \ Score = 2 \ * \frac{Recall \ * \ Precision}{Recall \ + \ Precision} \tag{10}$$

$$Accuarcy = \frac{TP \ + \ TN}{TP \ + \ TN \ + \ FP \ + \ FN} \ * \ 100 \tag{11}$$

Sensitivity also called Recall, is the true positive ratio (TP) correctly classified by diagnosis test. It describes how good the classifier is at classifying the correct tumor types. Specificity is the true negative ratio (TNR) of diagnostic test and shows how good the classifier is at predicting the negative condition. Precision is the positive predictive rate (PPR). F1-score measure the classification performance in term of recall and precision. Accuracy is the overall classification accuracy in term of TP and TN of the proposed method.

Table 7
Average five-fold overall classification performance of the proposed method on the test dataset.

Fine-tuning	Recall	Specificity	Precision	F1-Score	Accuracy
FT: B_1-B_6	94.25	94.69	89.52	91.73	94.82
FT: B_2-B_6	94.26	94.56	89.20	91.52	94.66
FT: B_3-B_6	94.04	94.21	88.23	90.93	94.26
FT: B_4-B_6	93.38	93.7	87.47	90.19	93.78
FT: B_5-B_6	91.50	91.85	83.99	87.40	91.92
FT: B_6	84.76	86.35	76.10	79.18	86.81

3.3. Results analysis and comparison

We evaluated the classification performance of the proposed model on five-fold cross-validation and summarized our results in the form of tables and graphical figures. The training-validation process of our best performing model with five-fold cross-validation is shown in Fig. 6. The benefit of transfer learning and fine-tuning is to reduce overfitting and speed the convergence. This benefit is very clear from the accuracy and the loss history of our proposed model. As shown in Fig. 6(a)–6(e), our proposed deep block-wise fine-tuned CNN reached its maximum performance very fast and there is a consistency between the training-validation accuracy and loss.

Another important aspect observed during experiments is the performance gap between shallow and deep fine-tuning. Earlier layers in CNN determines low-level features while later layers determine the domain-specific feature. Shallow fine-tuning is to fine-tune the last few fully connected layers only. Shallow fine-tuning achieved state-of-the-art results in computer vision applications because the nature of pre-trained CNN and fine-tune CNN datasets are similar and hence their low-level features are similar. Our dataset is medical brain MR images that are different from natural images. If we fine-tune only the last few layers then it will be difficult for CNN to learn relevant medical brain MR images features from natural images. This effect can be seen in shallow models FT: B_6 and FT: B_5-B_6. To achieve better performance, deep fine-tuning is required. Adding more blocks in fine-tuning gradually the performance increases gradually as shown in Table 5–7. Our experimental results show that classification performance increases gradually with incremental block-wise fine-tuning. Deep fine-tuned CNN learn low-level features of brain MR images in the early layers (as shown in Fig. 7) and higher layers learn domain-

Table 5
Average five-fold classification Sensitivity and Specificity for specific types of brain tumors of the proposed method on the test dataset.

Method	Sensitivity (TPR) or Recall			Specificity (TNR)		
	Glioma	Meningioma	Pituitary	Glioma	Meningioma	Pituitary
FT:B_1-B_6	95.97	89.98	96.81	93.79	96.42	93.93
FT: B_2-B_6	95.15	90.48	97.15	94.13	96.02	93.54
FT: B_3-B_6	94.20	91.06	96.87	94.19	95.25	93.19
FT: B_4-B_6	93.86	89.29	96.99	93.58	95.20	92.31
FT: B_5-B_6	91.70	86.71	96.09	91.97	93.53	90.04
FT: B_6	90.24	67.60	96.43	83.70	92.79	82.58

Table 6
Average five-fold classification Precision and F1-score for specific types of brain tumors of the proposed methods on the test dataset.

Method	Precision (PPR)			F1-score		
	Glioma	Meningioma	Pituitary	Glioma	Meningioma	Pituitary
FT:B_1-B_6	93.26	87.97	87.34	94.52	88.88	91.80
FT: B_2-B_6	93.64	87.14	86.82	94.29	88.60	91.66
FT: B_3-B_6	93.55	81.59	85.95	93.83	87.96	91.00
FT: B_4-B_6	92.91	84.82	84.67	93.30	86.88	90.38
FT: B_5-B_6	91.04	80.11	80.82	91.28	83.14	87.77
FT: B_6	83.44	74.23	59.83	86.48	75.03	81.42

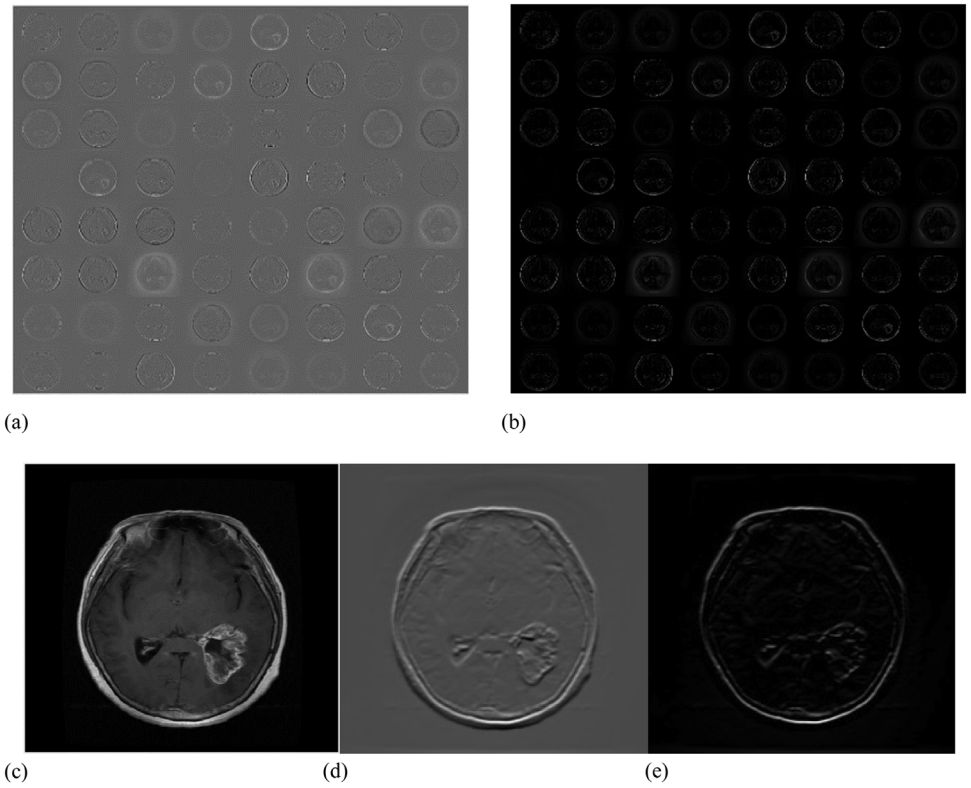


Fig. 7. Visual results of low-level features learned in the first block B_1 of fine-tuned model FT: B_1-B_6.

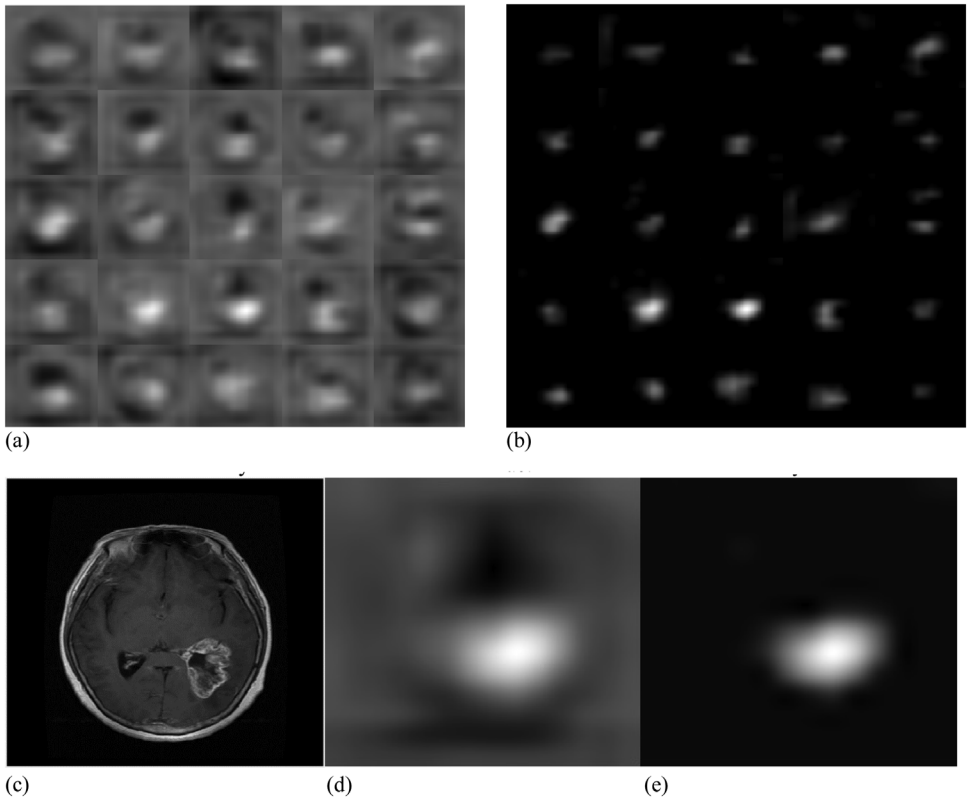
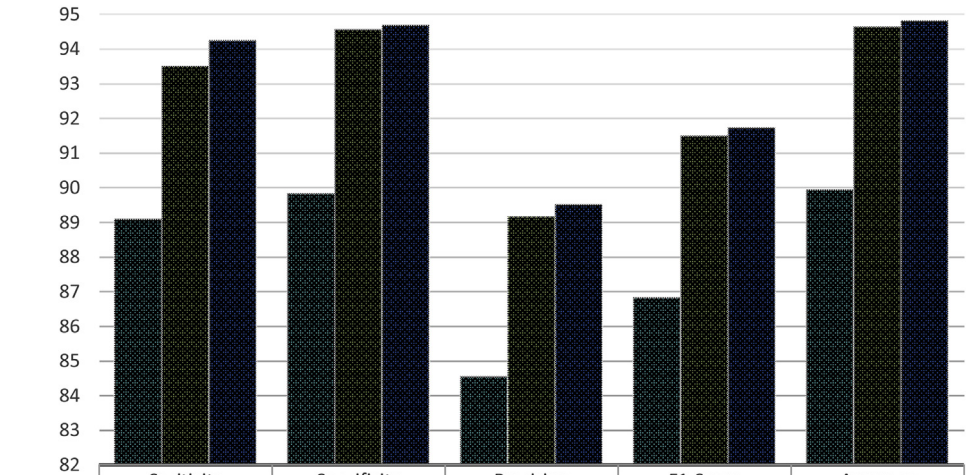


Fig. 8. Visual results of the high-level features in the fifth block of Fine-tuned model FT: B_1-B_6.

specific features of brain MR images (as shown in Fig. 8). We achieved the best performance from deep fine-tuned model FT: B_1-B_6.

Table 5 illustrates the average of five-fold classification sensitivity/recall and specificity, Table 6 presents precision and F1-score for individual/specific tumor type (i.e. Glioma, Meningioma, and



	Sesitivity	Specificity	Precision	F1-Score	Accuracy
AlexNet	89.095	89.84	84.56	86.83	89.95
VGG16	93.51	94.56	89.17	91.50	94.65
VGG19	94.25	94.69	89.52	91.73	94.82

Fig. 9. Results comparison of different pre-trained CNN models on CE-MRI dataset.

Table 8
Five-fold classification recall, Specificity, Precision, and F1-Score by proposed method on the test datasets.

Tumor Types	Test Set	Recall	Specificity	Precision	F1-Score
Glioma	Test set1	96.59	96.41	95.86	96.23
	Test set2	98.63	88.12	87.05	92.48
	Test set3	96.71	95.30	94.00	95.33
	Test set4	91.40	97.31	96.59	93.92
	Test set5	96.54	91.80	92.80	94.63
	Average	**95.97**	**93.79**	**93.26**	**94.52**
Meningioma	Test set1	93.01	97.66	93.01	93.01
	Test set2	82.14	96.91	89.03	85.45
	Test set3	92.98	96.65	87.60	90.21
	Test set4	93.48	94.95	84.31	88.66
	Test set5	88.28	95.95	85.91	87.07
	Average	**89.98**	**96.42**	**87.97**	**88.88**
Pituitary	Test set1	99.39	95.33	89.50	94.19
	Test set2	93.30	92.62	84.19	88.51
	Test set3	96.59	95.52	92.52	94.51
	Test set4	100.00	92.09	85.59	92.24
	Test set5	94.77	94.11	84.90	89.56
	Average	**96.81**	**93.93**	**87.34**	**91.80**

Pituitary) and Table 7 describes average overall classification performance of the proposed method on the test dataset. Table 8 shows the sensitivity/recall, specificity, precision, and F1-score of individual/specific tumor type and average of best performed proposed CNN model (FT: B_1-B_6) on five-fold testing datasets.

We also examined the transferability of knowledge from natural images to medical brain MR images. To observe the visual effect of the low-level and high-level features, we took feature maps from the deep fine-tuned model FT: B_1-B_6. B_1 describes the low-level features, while B_5 describes the high-level features. Fig. 7 visualizes the concept of low-level general features, while Fig. 8 visualizes the concept of high-level content specific features of MR images.

In Fig. 7(a) and 7(b), each square in the montage is the activation output of a channel in block B_1 (the second convolutional and the ReLu layer, respectively). In Fig. 7(a), white pixels represent strong positive activation, and black pixels represent strong negative activation. The channel with mostly gray does not activate strongly on the input image. The position of the pixel in the activation of a channel corresponds to the same position in the input image Fig. 7(c). The white pixel in a channel indicates that this position is strongly activated. When Fig. 7(d) is compared with the input image Fig. 7(c), it is clear that this channel activates on the edges. It activates positively on the light edges and negatively on the dark edges. However, only the positive activation is used, as shown in Fig. 7(b). If we compare the strongest activation channel of the convolutional and ReLu layers, as shown in Fig. 7(d) and 7(e) respectively, then Fig. 7(e) clearly identifies parts that have strong brain edge features.

In Fig. 8, the deep CNN learns domain (content) specific features in higher convolutional layers in a self-learning manner by combining features of earlier layers. We explored the fourth convolutional and ReLu layer in block B5 of the deep fine-tuned model (FT: B1-B6) in the same way as in Fig. 7. There are 512 feature maps, but for simplicity and page limitation, we explore only the 25 strongest activation channels, as shown in Fig. 8(a) and 8(b). These activation channels illustrate the interesting structure and focus on the brain area containing the tumor. Many channels in Fig. 8(a) contain areas of activation that are both light and dark. However, only the positive activation is used, as shown in Fig. 8(b). When the strongest activation channel of the convolutional layer in Fig. 8(d) is compared with the input image Fig. 8(c), it represents parts of the tumor structure. Similarly, if we compare Fig. 8(e) and 8(c), then Fig. 8(e) clearly shows that this strongest activation channel in the ReLu layer activates the tumor region. We did not provide the tumor information (such as tumor location, tumor segment or tumor boundary) to the CNN, but it has learned the tumor region to distinguish between classes of brain tumors in MR images. Conventional machine learning methods often use handcrafted features, but deep CNN can learn useful features by themselves.

We further investigated the usefulness of transfer learning and fine-tuning for smaller training datasets. For this purpose, we used one validation set for testing and the remaining four sets as training from the five-fold cross-validation datasets. We reduced the training data by randomly selecting 25, 50, and 75%. We trained FT: B_1-B_6 on the reduced training dataset. As described in Table 9, the deep fine-tuned VGG19 shows minor decrease in classification performance even with 25% training data. The relatively high performance by the deep fine-tuned CNN, even with smaller datasets, indicates the power of the feature representation and usefulness of transfer learning.

We also compared the classification performance of the three pre-trained CNN models (i.e., AlexNet, VGG16, and VGG19), trained

Table 9
Effect of reduction in training data on the performance of best performed deep fine-tuned CNN model (FT: B_1-B_6).

Training Data	Recall	Specificity	Precision	F1-Score	Accuracy
25%	91.37	91.69	84.03	87.33	91.75
50%	93.68	93.97	88.29	90.70	94.04
75%	94.75	94.77	89.30	91.83	94.74
100%	96.33	96.47	92.79	94.47	96.49

Table 10
Accuracy comparison among the proposed and state-of-the-art methods on the CE-MRI dataset.

Method	Cheng et al. (2015)	Paul et al. (2017)	Abiwinanda et al. (2019)	Proposed
Accuracy	91.28	90.26	84.18	94.82

on ImageNet (Russakovsky et al., 2015) dataset. VGG19 is 19 layers deep model that has been discussed in Section 2.4, AlexNet (Krizhevsky et al., 2012) is eight layers and VGG16 (Simonyan and Zisserman, 2014) is 16 layers deep CNN models. We applied the same transfer learning and fine-tuning strategy and performed the same experiments for AlexNet, VGG16, and VGG19 on CE-MRI dataset (Cheng, 2017). AlexNet is a shallow model as compared with VGG16 and VGG19. Therefore, AlexNet was fine-tuned in a layer-wise manner. AlexNet and VGG16 achieved the average five-fold classification accuracy of 89.95% and 94.65%, respectively. As shown in Fig. 9, VGG19 shows the best performance on all classification metrics among the three pre-trained models.

Table 10 describes the comparison of proposed classification with state-of-the-art methods on the same CE-MRI dataset. The classification results of the three compared methods are from the corresponding original papers. Cheng et al. (2015) used the BoW model and achieved the accuracy to 91.28%. Paul et al. (2017) adopted CNN and achieved the accuracy of 90.26%, while the Abiwinanda et al. (2019) introduced CNN and achieved the accuracy of 84.19%. Our proposed method achieved the highest classification accuracy of 94.82%. The MATLAB source code of the proposed research can be available on https://github.com/ZNKSWATI/BTC-MRI-TLFT.

4. Discussion

In this research, we focused on multiclass brain tumor classification for MR images using transfer learning and block-wise fine-tuning. The proposed transfer learning and block-wise fine-tuning can be used to train deep CNNs models for small datasets and may produce promising results, and the experiment results indicated that the performance of our proposed method is superior to that of state-of-the-art CNN based methods (Abiwinanda et al., 2019; Paul et al., 2017).

Training parameters are very important for CNN, but it's also time intensive procedure even on the high-speed GPUs. Therefore, we designed a new training manner to get sub-optimum parameters for all blocks, which ensured the fast and proper convergence. Another important aspect is the performance gap between shallow and deep fine-tuning. Earlier layers in CNN consist of low-level features while the later layers contain the domain-specific feature. Shallow fine-tuning is to fine-tune the last fully connected layers. Shallow fine-tuning achieved state-of-the-art performance in computer vision applications because the nature of datasets are similar, and hence their low-level features are similar. The CNN does not require to learn the same low-level features again, and if do so, then deep fine-tuned CNN will suffer from overfitting. Therefore, fine-tuning only for last a few layers can achieve enhanced performance in computer vision applications (Shin et al., 2016). However, CE-MRI dataset is different from natural images. If we fine-tune the last a few layers, then it will be difficult for the CNN model to learn relevant medical brain MRI features from natural images. To achieve better performance, deep fine-tuning is required. As gradually increasing the layers for fine-tuning, the performance increases gradually, as shown in Table 5–7. The proposed deep fine-tuned model learned low-level features of brain MR images in the earlier layers, as shown in Fig. 7. The higher layers learn domain-specific features of MR brain images from low-level features, as shown in Fig. 8. That is why we achieved the better performance from deep fine-tuning FT: B_1-B_6.

We adopted the VGG19 model because its deeper architecture can investigate the impact of deep fine-tuning and be suitable for feature representation in terms of localization or detection of specific content. We performed experiments on pre-trained AlexNet, VGG16, and VGG19. VGG19 achieved better performance than AlexNet and VGG16, as shown in Fig. 9.

The focus of current research is on brain tumor classification for MR 2D images, as CNN models are 2D in nature. We did not study the performance of fine-tuning in 3D MR images because the size of 3D medical datasets are large, for which full training of CNNs from scratch had shown promising performance. Our research is based on 2D images, so, 3D pre-trained CNN models are not applicable to the 2D CE-MRI dataset. Ultimately, pre-trained VGG19 is the natural choice for 2D brain tumor MR images classification to investigative the transfer learning and block-wise fine-tuning on CE-MRI dataset.

5. Conclusion and future work

In this paper, we proposed a new approach for brain tumor image classification based on transfer learning and fine-tuning. The proposed strategy of transfer learning with block-wise fine-tuning suggests an alternative approach, which is different from using pre-trained CNN as an off-the-shelf feature extractor (without training) that train the separate method for classification (such as k-nearest-neighbors, Support Vector Machines, Boosted Trees, Decision Trees, and Random Forest). It also demonstrates the transferability of learning from natural images to medical brain MR images. The proposed approach may be used to develop classification system for other body organ MRI images and other medical imaging domains, such as X-rays, PET, and CT. Our method is more generic because it requires minimum preprocessing for 2D MR images and does not use handcrafted features. The experimental results shown that the proposed method not only outperformed state-of-the-art traditional machine learning method but also outperformed state-of-the-art CNNs methods on the same CE-MRI dataset. In future work, we will add normal brain CE-MR Images to the dataset, which may aid to differentiate tumors classification further.

Conflict of interest

The authors declare no conflict of interest.

Acknowledgment

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